

Model-based evaluation of synthetic financial time series data: A comparative study with multi-metric validation

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ABSTRACT

This research presents a behaviorally informed framework for synthesizing financial time-series data, specifically designed to emulate the complex dynamics of foreign exchange markets. Deviating from conventional generative adversarial networks (GANs) or purely statistical distribution-matching, the proposed methodology adopts a game-theoretic architecture. This framework integrates trader-interaction dynamics, stochastic strategies, and information asymmetry, treating the market as a strategic participant to reproduce authentic volatility patterns and structural dependencies. To ensure numerical stability across extensive simulations, the study introduces a uniform upward scaling procedure and controlled initialization, preventing pathological price behaviors without compromising the underlying statistical properties. The framework's analytical fidelity was rigorously evaluated against a suite of econometric and machine learning models, including ARIMA, XGBoost, LSTM, N-BEATS, and DLinear. Experimental results involving 12,960 hourly observations demonstrate that the synthetic data maintains strong alignment with empirical benchmarks. DLinear emerged as the superior model, exhibiting exceptional stability with an R^2 frequently exceeding 0.98 and a Mean Absolute Scaled Error (MASE) near unity. While XGBoost and N-BEATS yielded competitive results, ARIMA and LSTM showed anticipated performance degradation due to temporal noise. Comprehensive residual diagnostics, including Ljung-Box tests and stationarity assessments, confirm that the generated series are behaviorally consistent and analytically reliable. This framework thus provides a robust foundation for comparative modeling and experimental financial research.

1. Introduction

Demand for predictive accuracy in financial markets is on the rise yet access to clean and comprehensive datasets is increasingly being constrained by the requirements of data privacy regulations [1]. Forex trading, in particular, is sensitive to data disclosure due to its real-time and proprietary nature [2]. indicates that financial institutions develop internal policies to protect the important relationship of trust with their customers. For example, most institutions enforce strict data-governance rules, role-based access controls that ensure employees only view data relevant to their duties, and customer-privacy policies that prohibit external sharing without consent. Additional internal frameworks, such as information-security procedures, vendor-management requirements, and 'Chinese-wall' policies separating business units, further reduce how sensitive financial datasets can be used. Organizations across the globe have to ensure compliance with the various regulations pertaining to privacy and legal aspects of

customer data. These challenges underline the critical need for innovative approaches, such as the generation and validation of synthetic datasets, to augment and diversify data for financial forecasting [3]. Data generation offers a way to create datasets that mimic the statistical properties of real-world data while adhering to required regulations. Nonetheless, the validity of synthetic data for predictive financial modeling remains a critical concern [4]. Validation ensures that a model or method is generating correct data, which portrays the expected financial data features and is useful in prediction of market trends. Besides, generated data is verified to ensure it aligns with real-world data observations and remain consistent in its performance.

Financial time series modeling, particularly within forex markets, presents persistent challenges arising from volatility clustering, pronounced non-stationarity, regime shifts, and the structural complexity inherent in multi-scale market dynamics. Empirical behaviors of asset prices such as heavy-tailed return distributions, volatility clustering, non-stationarity, limited linear autocorrelation in returns, and regime-

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dependent volatility emerge consistently across major financial markets and asset classes. These behaviors exhibit remarkably similar patterns regardless of market or instrument, indicating that the underlying statistical and behavioral regularities, commonly referred to as stylized facts [5], are not market-specific anomalies but rather universal properties of speculative dynamics [6]. Consequently, methodologies and empirical observations from equities, commodities, and cryptocurrencies are relevant to foreign exchange analysis, as these markets share time-series characteristics such as volatility clustering, non-stationarity, and heavy tails. In this work, volatility clustering is modeled through the structured exploitation of static stop-loss strategies by traders, while non-stationarity and heavy tails are captured via asymmetry, stochastic trader behavior, and the volatility-engineering process. Beyond descriptive statistics based on stylized facts, predictive-modeling studies demonstrate cross-domain consistency in performance, model stability, and error sensitivity. These comparative insights from broader financial contexts provide a solid theoretical foundation for analyzing and interpreting forex dynamics.

As noted by Refs. [5,7], financial time series exhibit well-documented statistical regularities often termed stylized facts that characterize real market behavior. These include non-stationarity, where price levels follow a unit-root process $x_t = \rho x_{t-1} + \epsilon_t$; volatility clustering, captured by conditional heteroskedasticity in a GARCH (1,1) form $\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \beta_1 \sigma_{t-1}^2$; and heavy-tailed return distributions, reflected by high kurtosis $K = E[(r_t - \mu)^4] / (\sigma^4)$. These features are critical for assessing the realism of synthetic financial data, as they determine whether simulated markets reproduce the persistence, asymmetry, and scale-dependent variability observed in empirical series. Formal statistical definitions, test formulations, and corresponding diagnostics are presented in Section 3.2.

Further to the stylized facts examination, we assess whether synthetic data preserves the distributional shape, structural breaks, and non-linear dependencies observed in real financial series. These attributes are critical for downstream applications such as regime clustering, anomaly detection, and risk assessment. Among the available similarity measures, the Wasserstein distance, grounded in optimal transport theory, provides a robust means of quantifying how much “effort” is required to transform one probability distribution into another [8]. For one-dimensional distributions P and Q with cumulative distribution functions $F_P(x)$ and $F_Q(x)$, the first-order Wasserstein distance is defined in (1):

$$W_1(P, Q) = \int_{-\infty}^{\infty} |F_P(x) - F_Q(x)| dx \quad (1)$$

This property makes it particularly well suited for comparing return distributions that exhibit heavy tails, asymmetry, or other deviations from normality. We use it to evaluate the fidelity of synthetic datasets, ensuring that critical statistical properties of real financial returns are accurately reproduced for predictive modeling.

Complementing this, the Hausdorff distance provides a geometric measure of similarity between the trajectories of real and synthetic series [9]. For two sets of points $A = \{a_i\}$ and $B = \{b_j\}$ the Hausdorff distance is expressed in Equation (2):

$$H(A, B) = \max \left\{ \sup_{a \in A} \inf_{b \in B} d(a, b), \sup_{b \in B} \inf_{a \in A} d(a, b) \right\} \quad (2)$$

The Hausdorff distance measures the greatest distance from any point in one set to the nearest point in the other, effectively capturing localized structural differences or misalignments in time-dependent behavior. In this work, we apply it to financial time series to evaluate how closely the synthetic data replicates the temporal geometry and continuity of the real series, providing a complementary perspective to purely distributional comparisons.

In classical time series analysis, a set of statistical methods are used to understand, model, and forecast time-dependent data. Central to this

approach are techniques like Autoregressive Integrated Moving Average (ARIMA), which model data based on its own past values, trends, and noise components [10]. These models assume linearity and stationarity, making them particularly effective for well-behaved, regularly patterned datasets. Classical time series methods emphasize interpretability and theoretical rigor, providing valuable insights into the structure and behavior of temporal data. Despite their limitations in handling non-linear or complex patterns, these techniques remain foundational in time series forecasting and serve as important benchmarks for evaluating more advanced models. While classical approaches rely on pre-defined linear structures, the application of machine learning (ML) models in financial forecasting has gained substantial attention for their ability to capture intricate, nonlinear patterns and dependencies in data [11]. These models learn representations directly from complex data, enabling adaptive and automated prediction beyond the scope of traditional methods. In our study, ARIMA models provide a transparent benchmark for evaluating whether synthetic forex data preserves linear dependencies, trends, and noise structures, offering a foundational reference point against which more complex models can be assessed.

Long Short-Term Memory (LSTM) networks have emerged as a particularly effective tool for time-series prediction, excelling in domains such as forex trading, where market dynamics are inherently sequential [12]. These values can be represented as a sequence of data points indexed in time order, often modeled as a function or a sequence of observations $y_1, y_2, y_3 \dots y_t$ where each y_t corresponds to a value observed at time t . LSTMs are specialized form of Recurrent Neural Networks (RNNs) uniquely equipped to address challenges of sequential data processing through their ability to retain long-term dependencies [13]. Each LSTM unit comprises memory cells that store data streams across time steps, with an internal mechanism that regulates the addition, retention, or removal of information enabling the seamless transfer of relevant data by linking the past, present and future. Furthermore, the independence of individual memory cells empowers LSTMs to effectively filter, update, and propagate values, ensuring adaptive learning within complex temporal sequences [14]. LSTMs are employed in our work to assess the ability of synthetic data to retain sequential dependencies and long-term patterns characteristic of real forex time series, capturing the complex temporal dynamics that classical models may overlook.

Ensemble ML models have gained prominence due to their ability to combine the strengths of multiple learners, thereby improving predictive accuracy and generalizability [15]. By aggregating the outputs of several base models, often decision trees, ensemble methods reduce the risk of overfitting and capture complex, non-linear patterns in data. Among these, Extreme Gradient Boosting (XGBoost) has emerged as a leading technique, offering high performance and efficiency through optimized gradient boosting algorithms [16]. XGBoost incorporates regularization, parallel processing, and tree pruning, making it particularly robust in handling noisy or imbalanced datasets. Its widespread success across domains such as finance [17], bioinformatics [18], and natural language processing [19] demonstrates its versatility and effectiveness. Consequently, XGBoost serves as a strong benchmark in forecasting tasks where both accuracy and computational efficiency are critical. In our analysis, XGBoost serves as a high-performance, non-linear benchmark, enabling evaluation of how well synthetic forex data supports accurate predictions in scenarios with complex interactions and noisy market conditions.

Neural Basis Expansion Analysis for Interpretable Time Series Forecasting (N-BEATS) represents a significant advancement in deep learning-based forecasting architectures. Unlike traditional recurrent or convolutional models, N-BEATS is a fully connected feed-forward neural network that employs backward and forward residual links to capture temporal dependencies without explicit sequence modeling. The architecture decomposes a time series into interpretable components such as trend and seasonality through its basis expansion layers, enabling both high predictive accuracy and model transparency [20]. Its design

eliminates the need for domain-specific feature engineering, allowing it to learn directly from raw historical data. Empirical studies have demonstrated that N-BEATS and its exogenous extension (N-BEATSx) consistently outperform classical statistical models and competing deep learning approaches across diverse forecasting benchmarks, including electricity load and price forecasting [21]. Owing to its modular structure, scalability, and interpretability, N-BEATS serves as a robust benchmark for evaluating the predictive fidelity of synthetic data in contexts demanding both accuracy and explainability. We leverage N-BEATS to test the synthetic data's capacity to preserve interpretable temporal structures, such as trend and seasonality, allowing us to examine both predictive accuracy and transparency in forecasting applications.

Decomposition-Linear (DLinear) represents a notable advancement in deep learning based time series forecasting by emphasizing simplicity and efficiency. Unlike complex recurrent or attention-based architectures, DLinear is built upon a fully linear model that directly predicts future values by decomposing the time series into trend and seasonal components through a simple decomposition module [22]. This approach captures temporal dependencies without requiring explicit sequence modeling or deep nonlinear transformations, thereby reducing overfitting and computational overhead. By learning directly from raw historical data, DLinear eliminates the need for extensive feature engineering while maintaining strong predictive performance. Empirical evaluations have shown that DLinear consistently outperforms both classical statistical methods and more sophisticated deep learning models, particularly for long-term forecasting tasks across various benchmarks, including electricity load and traffic forecasting. Its straightforward architecture, interpretability, and computational efficiency make DLinear a compelling baseline for assessing the predictive fidelity of synthetic time series data in scenarios that demand both accuracy and scalability [23]. DLinear is applied as a computationally efficient and interpretable deep learning baseline, allowing us to evaluate the fidelity of synthetic data in capturing decomposable trend and seasonal components while minimizing overfitting and computational overhead.

To enhance predictive accuracy for ML models deployed in time series forecasting, we focus primarily on minimizing of error metrics. This optimization process is driven by a loss function, which quantifies the discrepancy between predicted and actual values. The model iteratively updates its parameters to reduce this loss, thereby improving forecasting performance. Recent literature in forecasting and time series analysis underscores the critical role of performance metrics in robust model evaluation and benchmarking, particularly in applied domains such as energy [24], finance, and economics [25], where various regression loss functions are commonly employed. In this study, we compute and report the mean absolute error (MAE), mean squared error (MSE), coefficient of determination (R^2), mean absolute percentage error (MAPE), symmetric mean absolute percentage error (SMAPE) and symmetric absolute scaled error (MASE) for models trained independently on both real and synthetic datasets.

Following the approach of [26,27], we evaluate models trained on both real and synthetic datasets using standard statistical measures: Mean Absolute Error (MAE), Mean Squared Error (MSE), Coefficient of Determination (R^2), Mean Absolute Percentage Error (MAPE), Symmetric Mean Absolute Percentage Error (SMAPE), and Mean Absolute Scaled Error (MASE). Mathematical definitions and interpretive significance of each metric are provided in Refs. [26,27], ensuring reproducibility and clarity.

MAE measures the average magnitude of prediction errors, reflecting model accuracy and precision. MSE capture the average squared deviations, penalizing larger errors more heavily and thus indicating predictive reliability. R^2 quantifies the proportion of variance in the target variable explained by the model, serving as a scale-independent measure of goodness-of-fit.

MAPE expresses forecast error as a percentage of actual values,

enabling interpretability across different data scales. SMAPE refines MAPE by symmetrically normalizing errors over the mean of actual and predicted values, thereby reducing bias between over and under-predictions. MASE compares a model's mean absolute error to that of a naïve benchmark forecast, yielding a scale-independent indicator of relative performance, where values below one denote superior predictive capability. SMAPE and MASE are computed as shown in Equations (3) and (4), respectively:

$$SMAPE = \frac{100}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2} \quad (3)$$

$$MASE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{\frac{1}{n-1} \sum_{j=2}^n |y_i - y_{j-1}|} \quad (4)$$

Collectively, these metrics offer a comprehensive framework for assessing model accuracy, robustness, and comparability across heterogeneous time series datasets.

1.1. Research gap

Despite substantial progress in ML-based financial forecasting and synthetic data generation, a clear gap remains in evaluating the predictive utility of synthetic forex data. Prior studies typically decouple synthetic data generation from downstream forecasting performance, providing limited insight into how data fidelity affects model accuracy. In particular, few works systematically assess synthetic OHLC forex data across classical, linear, deep learning, and ensemble models, and forecast error behavior, especially scale-sensitive metrics such as MASE is rarely examined in synthetic forex settings. While GAN-based approaches have been explored for financial time series generation, challenges such as variable ability to capture stylized facts across architectures and issues like mode collapse and unstable training have limited their effectiveness in producing synthetic sequences suitable for comprehensive forecasting evaluation [28].

Moreover, most existing approaches prioritize statistical or correlation-based similarity, often overlooking whether synthetic data preserves market mechanisms or behavioral structure. As a result, synthetic sequences may match marginal distributions while lacking economically meaningful dynamics, particularly those arising from trader-market interactions.

Prior studies in finance and adjacent fields [29–33] largely concentrate on statistical fidelity in isolation. In contrast, only limited research evaluates stylized facts, distributional characteristics, and the extent to which forecasting performance transfers between real and synthetic forex data. Our work addresses this gap through a unified evaluation framework.

1.2. Contribution and novelty

This study proposes a unified framework for generating and validating synthetic OHLC forex data that integrates stochastic game-theoretic data generation with model-based predictive evaluation. Unlike prior work that emphasizes either similarity metrics or forecasting accuracy in isolation, the framework embeds trader-market interactions into the generative process and evaluates the resulting data across linear (DLinear), classical (ARIMA), ensemble (XGBoost), and deep learning models (LSTM, N-BEATS).

The contribution of this study is threefold. First, it demonstrates that synthetic data can be meaningfully evaluated through predictive transferability, rather than relying solely on statistical resemblance. Second, it introduces a multi-metric validation strategy that combines stylized facts, distributional distance, and forecast error behavior to assess both structural and predictive fidelity. Finally, by incorporating DLinear, the study provides an explicit linear baseline that helps

disentangle trend-seasonal effects from nonlinear learning, thereby strengthening the interpretability of synthetic data performance.

Overall, the framework offers a replicable and balanced benchmark for assessing high-fidelity synthetic financial data, supporting both robust forecasting and principled evaluation across econometric and machine learning paradigms.

2. Related work

This section reviews prior studies that contribute to the understanding of the need for synthetic data generation, predictive modeling, and the validation of synthetic datasets in the financial domain.

In the study by Ref. [34], it has been observed that financial data often contains highly sensitive and personally identifiable customer information significantly restricting its use outside organization that generates it which subsequently limit its use. Generative models such as GANs and agent-based approaches have been widely explored for synthetic financial data creation. While GAN-based methods reproduce statistical distributions from real data [33–35], agent-based simulations capture underlying market mechanisms [31,36]. However, both approaches face validation challenges, GANs risk overfitting, and ABMs can be complex to calibrate, motivating hybrid frameworks that balance behavioral realism with statistical fidelity. Building on this foundation, this study enhances data generation process and rigorously validates the synthetic data for its suitability in predictive modeling tasks.

Empirical findings in financial econometrics have long informed the modeling of market dynamics through a set of well-documented stylized facts most notably non-stationarity, volatility clustering, and excess kurtosis and leverage effect. Non-stationarity and the persistence of Cont's original stylized facts have been reaffirmed in modern markets [35]. Volatility clustering, evident in significant autocorrelations of absolute returns across multiple lags, is particularly pronounced in USD/EUR (i.e., EUR/USD) daily returns from 1999 to 2022, demonstrating long memory and time-varying volatility [36]. Likewise, agent-based financial models since 2001 have consistently reproduced excess kurtosis in simulated returns, capturing the heavy-tailed distributions observed in empirical data [7]. The leverage effect, describing the negative correlation between returns and subsequent volatility, remains well established in equities but weak or statistically indistinct in foreign exchange markets, where asymmetry depends on the modeling framework and sample period [37,38]. Collectively, these four properties, non-stationarity, volatility clustering, excess kurtosis, and the leverage effect serve as essential benchmarks for assessing the realism of both real and synthetically generated forex data. Accordingly, in this study, their consistency across real and synthetic series is treated as a primary indicator of representational fidelity and suitability for downstream predictive modeling.

Recent work has made significant strides in developing rigorous validation methodologies for synthetic and real time-series data, especially in financial and generative contexts [39]. proposed a projected Wasserstein distance two-sample test that mitigates the curse of dimensionality by finding low-dimensional linear projections that maximize discrepancy between empirical distributions; they also provide theoretical finite-sample convergence guarantees and show strong performance on synthetic vs. real data comparisons [40]. conducted a large-scale study of utility metrics for synthetic data generation, validating various metrics (including distance/divergence measures and

classifier-based distinguishability) and demonstrating how they effectively rank different generative models based on downstream predictive workload performance. Much of the recent literature has focused on adapting or extending Wasserstein distance for finance. For Hausdorff distance, though less directly in forex, there are works comparing Hausdorff and related distances in tasks of detecting change points or clustering change-point structures. For example [9], compares Hausdorff distance and Wasserstein among others in clustering time series' structural breaks while [41] introduces a version of Hausdorff distance adapted to multivariate time series generation, improving over standard versions in handling temporal structure.

Recent research in synthetic data evaluation has increasingly emphasized similarity-based metrics as a robust mechanism for quantifying the fidelity of generated datasets [42]. propose a comprehensive framework that integrates multiple complementary metrics: the Wasserstein-Cramér's V test assesses distributional similarity between real and synthetic data, the Novelty Test detects structural overlap or unique patterns, the Domain Classifier Test evaluates whether a ML model can distinguish synthetic from real samples, and the Anomaly Detection Test examines internal consistency and the presence of realistic variability. By combining these measures, the framework provides a multidimensional assessment that ensures synthetic data not only replicates statistical distributions but also maintains practical utility for downstream analytical and predictive tasks [43]. offer a comprehensive survey of time-series anomaly detection methods including evaluation protocols and metrics, which are highly relevant for "novelty test" style validation in forex time series synthetic data. Together, these evaluations provide a multi-faceted empirical foundation for validating synthetic data in time-series domains like forex, supporting tests that incorporate domain classifier separation, distributional similarity and novelty/anomaly detection.

Recent studies demonstrate that deep learning models, especially LSTMs, outperform classical approaches such as ARIMA in forex forecasting due to their ability to capture non-linear dependencies [44,45]. Ensemble methods like XGBoost further improve robustness and generalizability [46,47]. More recently [22,23], linear deep models such as DLinear have been shown to provide competitive performance by explicitly decomposing trend and seasonal components, offering a transparent baseline that isolates linear structure from nonlinear learning. Together, these models establish complementary empirical baselines for assessing the predictive equivalence and transferability of synthetic versus real forex datasets.

Recent work emphasizes residual diagnostics as essential for validating hybrid and deep-learning time-series models. In Ref. [48], Augmented Dickey–Fuller (ADF) and Ljung-Box tests assessed residual stationarity and whiteness in a hybrid ARIMA-LSTM framework. Similarly [49], applied Autocorrelation Function (ACF) and Ljung-Box diagnostics in a CEEMDAN-SE + ARIMA-CNN-LSTM model for financial forecasting. Further [50], embedded the Ljung-Box statistic within a hybrid model's loss function to suppress residual autocorrelation. These studies collectively affirm that applying ADF, ACF, and Ljung-Box tests ensures residual independence and stationarity, thereby enhancing the interpretability and statistical soundness of deep-learning forecasting models.

In their studies [51], claims that ensuring synthetic data accurately reflects real-world distributions is a vital step in establishing its credibility for predictive modeling and observes that quantitative analysis is

generally quite difficult to perform on generative models, especially in the context of this relatively new issue of data generation. In some other studies by Ref. [5], synthetic financial data is assessed using divergence metrics for distributional similarity and further tests the data's utility in downstream tasks like predictive modeling. Our study first evaluates the consistency of stylized facts across real and synthetic datasets, then advances to comparative model assessment using metrics suited to scenarios involving non-mapped data with differing scales.

Even though financial time series forecasting is a subset of time-series studies, due to the embedded profit-making expectations from successful prediction models, some differences exist, such that higher prediction accuracy sometimes might not reflect a profitable model [52]. This underscores the critical need for risk management. In contemporary trader education and algorithmic strategy design, risk management frameworks prioritize the implementation of stop-loss orders as an essential mechanism to cap downside exposure and mitigate significant financial losses. Studies by Ref. [53] rigorously analyze optimal trading strategies incorporating stop-loss constraints, demonstrating how these orders serve as critical safeguards in the presence of market uncertainty and transaction costs. Their findings emphasize that stop-loss exits are not merely tactical choices but foundational to disciplined trading systems that seek to manage risk proactively. Complementing this [54], highlights stop-loss orders as fundamental tools embedded within algorithmic trading protocols, underscoring their role in enforcing predefined risk thresholds and preserving capital integrity. Collectively, these studies reflect a consensus in both academic and practitioner communities that stop-loss orders form a non-negotiable baseline in risk management, often prioritized over other trade exit mechanisms such as take-profit targets, which are typically treated as discretionary or secondary components within trading strategies.

While [55] associates volatility clustering to various factors including the information-processing behavior of market participants, investor sentiment, and structural market dynamics, our approach extends this understanding. Theoretically, we demonstrate how stop-loss dynamics can precipitate abrupt price movements and reversals, thereby reproducing volatility clustering within synthetic data. In this setup, the market behaves as an intelligent and strategic agent that learns to exploit predictable trader behavior, profiting from liquidity concentrations around stop-loss zones and reinforcing the adaptive, self-organizing character of real-world financial markets.

3. Methodology

This section details the methodology for generating and validating synthetic financial time series. Key stylized facts such as volatility clustering, non-stationarity, lagged effects, and kurtosis are examined in both real and synthetic datasets. Synthetic data fidelity is assessed using distance and similarity metrics, followed by independent training and evaluation of ML models on each dataset.

3.1. Data description

This study employs two datasets: real forex data and synthetic forex data. The real EUR/USD historical data was obtained from two sources, ForexSB [56], covering the period from September 21st, 2009 to October 1st, 2025, representing live trading streams, and the Kaggle repository [57](EURUSD_1H_2020-2024.csv). From each source, three subsets consisting of 4,320, 8640 and 12960 consecutive 1 h entries were extracted. We generate synthetic data by building upon the methodology proposed by Ref. [58]. Using both real datasets ensures evaluation across commonly referenced historical data and authentic live-market dynamics.

We introduce an enhanced framework to generate synthetic data by incorporating boundary management for prices approaching or crossing zero, model major moves and reversals to represent volatility clustering, and adjust the scaling factor. Collectively, these refinements contribute

to improved data fidelity and closer alignment with empirical market behavior. The subsequent paragraph outlines the methodology behind these improvements in synthetic data generation.

In the methodology by Ref. [58], the market is conceptualized as a strategically active participant (an “intelligent market”) operating within a stochastic, imperfect-information environment, rather than as a neutral entity. The framework involves sequential interactions, where the market alternates with traders: traders have limited visibility of the market's actions, while the market, with full information, counteracts each trader's strategy. To clarify, this construct is a mathematical abstraction intended to emulate liquidity-hunting dynamics within a controlled simulation, rather than implying deliberate manipulation in real financial markets. Allowing unrestricted initial price choice in Ref. [58], combined with randomized trade types and risk appetites, can lead to data-induced instability, particularly when prices approach zero or negative values, potentially causing numerical problems in downstream analyses, an issue not addressed in prior work. Building upon this framework, our methodology incorporates adjustments to ensure prices remain positive, thereby enhancing numerical stability in the forex market model. To address this instability, while a permanent safeguard against extreme scenarios in large-scale simulations cannot be guaranteed, initializing the price above 2.00000 consistently prevented it from reaching zero or negative values in our tests, which included up to 46,668,960 observations. A more general solution involves uniformly scaling generated data upward during preprocessing, effectively eliminating zero or negative values without altering the overall distribution. Trader rewards are calculated based on changes in the exchange ratio between entry and exit positions, while the market's inversely defined reward enforces systemic balance through adaptive feedback, preserving stability and behavioral realism across simulations.

We further enhance the generation procedure by incorporating sudden price moves and reversals to emulate engineered volatility. As illustrated in Fig. 1, traders can be systematically exploited when stop-loss (SL) mechanisms are active. Specifically, we model a scenario with two traders: trader X initiates a long (buy) position, followed by trader Y taking a short (sell) position, both employing stop-loss orders to manage downside risk. The intelligent market strategically induces bidirectional price swings that intersect static SL levels, triggering forced exits and resulting in systematic trader losses. To replicate this behavior, the simulation introduces high-volatility events, sharp price movements occurring at least once every 36-h interval, followed by mean-reverting corrections that restore the exchange rate to its prior range. Although explicit SL attributes are not defined in the simulation, this mechanism effectively produces volatility clustering and enhances the intelligent market's reward dynamics without altering the core model design.

The scaling reduction from 0.001 in Ref. [58] to 0.00025 (i.e., 75 %

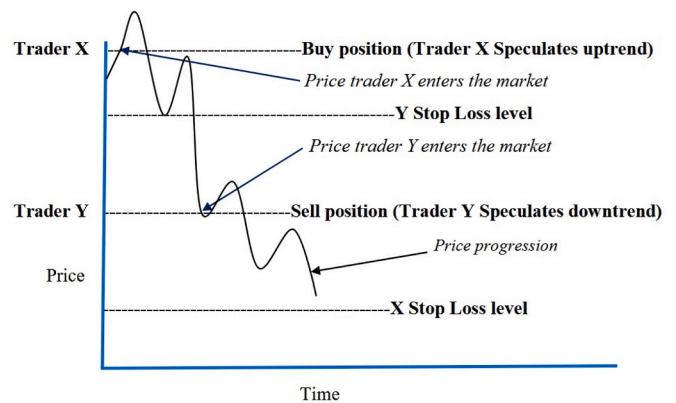


Fig. 1. Simulation of engineered volatility events demonstrating how the intelligent market exploits trader stop-loss mechanisms through induced price swings and mean-reverting corrections.

decrease) is introduced as a regularization-like adjustment to mitigate excessive price jumps that would otherwise distort the simulated dynamics. Although not directly derived from prior literature, the coefficient was empirically calibrated through sensitivity analysis to preserve volatility clustering while maintaining numerical stability, analogous to stabilization techniques employed in generative models such as Quant GANs [59]. The coefficient was selected through a straightforward sensitivity analysis in which the scaling parameter was varied over the range 0.0001–0.001 in increments of 0.00005. For each value, we evaluated, stability of the ratio update process (no runaway divergence), preservation of volatility clustering measured via rolling variance and

autocorrelation of squared returns, and absence of unrealistically large price jumps exceeding twice the empirical average of real data. The value 0.00025 was chosen because it consistently satisfied all three conditions across multiple simulation runs.

Pseudocode 1: Forex_Game_Simulator Algorithm.

To model the stochastic interaction between traders and the market, **Pseudocode 1** (Forex_Game_Simulator) simulates a sequence of trades characterized by dynamic ratio adjustments and alternating risk phases. The algorithm generates trade directions, associated risk appetites, and evolving ratio values across a specified number of trades. High-risk phases are introduced at random intervals, during which the ratio

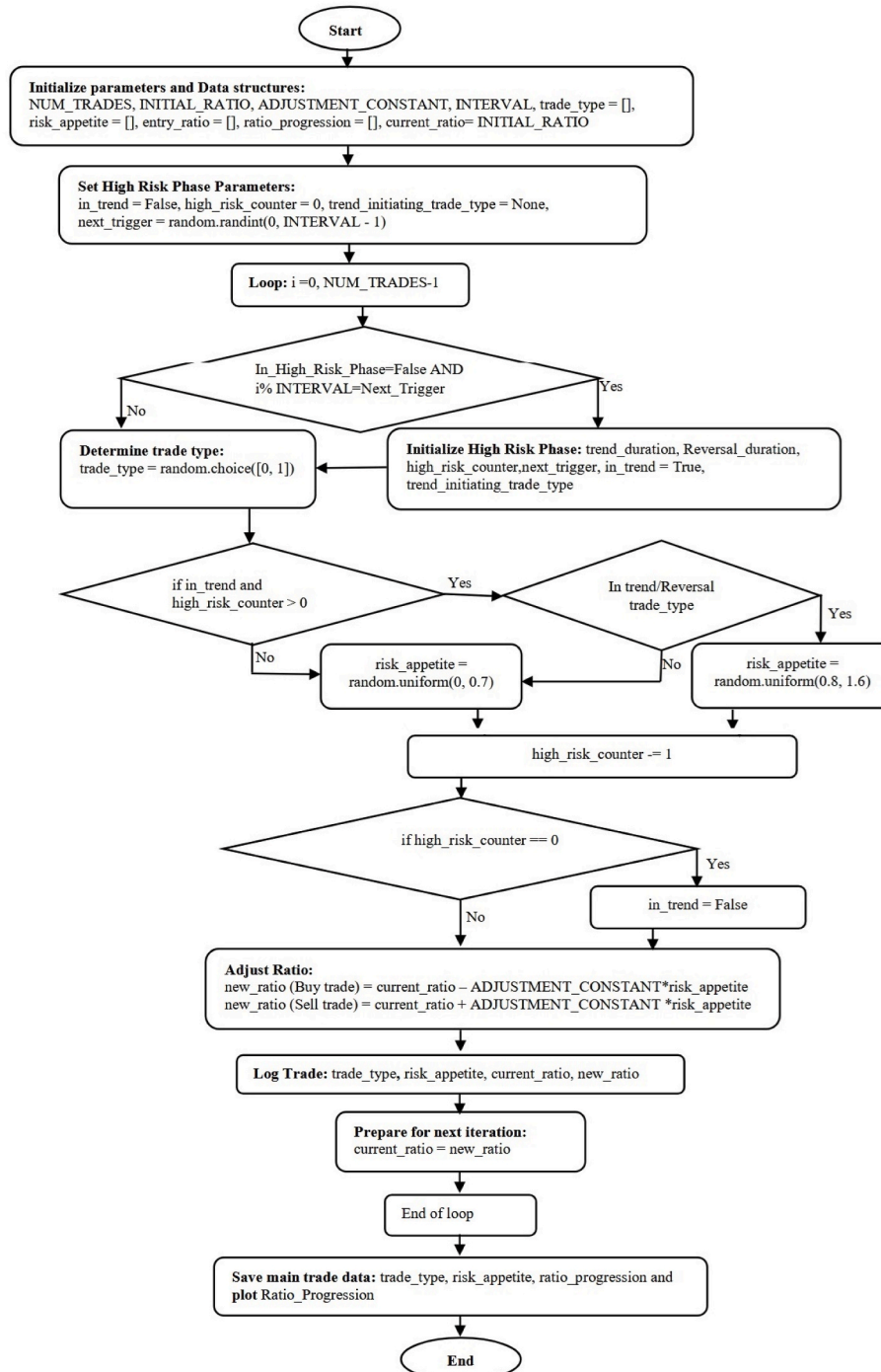


Fig. 2. Flowchart of the Forex Game Simulator illustrating trade generation, high-risk phase activation, risk assignment, ratio updates, and state transitions across simulated trades.

experiences amplified fluctuations depending on trade type and risk intensity. The resulting sequences form the foundation for analyzing emergent market behaviors under controlled stochastic conditions. See Fig. 2 for the flowchart illustrating **Pseudocode 1**.

Pseudocode 1

Forex_Game_Simulator

Input: NUM_TRADES, INITIAL_RATIO, ADJUSTMENT_CONSTANT, INTERVAL
Output: ratio_progression[], trade_type[], risk_appetite[]

```

1  current_ratio ← INITIAL_RATIO
2  Initialize: trade_type[], risk_appetite[], entry_ratio[], ratio_progression[]
3  in_high_risk_phase ← False
4  high_risk_counter ← 0
5  high_phase_DirectionA_length ← 0
6  high_phase_DirectionA_Reversal_length ← 0
7  initiating_trade_type ← None
8  next_trigger ← UniformInteger(0, INTERVAL - 1)
9  For i = 0 to NUM_TRADES - 1 do
10   If (in_high_risk_phase = False) and (i mod INTERVAL = next_trigger) then
11     Initialize high-risk phase parameters
12     initiating_trade_type ← UniformChoice({0,1}) # Buy = 0, Sell = 1
13     in_high_risk_phase ← True
14     Reset high_risk_counter and phase lengths
15     next_trigger ← UniformInteger(0, INTERVAL - 1)
16   End If
17   trade_type_i ← UniformChoice({0,1})
18   If in_high_risk_phase = True and high_risk_counter > 0 then
19     If (alignment and reversal conditions satisfied) then
20       risk_appetite_i ← Uniform(0.8, 1.6)
21     Else
22       risk_appetite_i ← Uniform(0, 0.7)
23     End If
24   Else
25     risk_appetite_i ← Uniform(0, 0.7)
26   End If
27   price_adjustment ← ADJUSTMENT_CONSTANT × risk_appetite_i
28   If trade_type_i = 0 then # Buy
29     new_ratio ← current_ratio - price_adjustment
30   Else # Sell
31     new_ratio ← current_ratio + price_adjustment
32   End If
33   Record trade_type_i, risk_appetite_i, current_ratio, new_ratio
34   Append new_ratio to ratio_progression
35   If in_high_risk_phase = True then
36     high_risk_counter ← high_risk_counter - 1
37     If high_risk_counter = 0 then
38       in_high_risk_phase ← False
39     End If
40   End If
41   current_ratio ← new_ratio
42 End For
43 Return ratio_progression, trade_type, risk_appetite

```

Using **Pseudocode 1**, we generate three sets of data containing 15,556,320, 31,112,640, and 46,668,960 observations. These observations constitute sequential prices at assumed interval of 1 s. We cluster the prices into non-overlapping intervals of 3601 consecutive price (1 Hour), noteworthy, first second is the period between observation 0 and 1. Within each interval, the first price is designated as Open, the highest price above the Open is labeled as High, the lowest price below the Open is identified as Low, and the last price is marked as Close. The Open, High, Low, and Close (OHLC) values for each interval are extracted sequentially and organized into a tabular format. When representing data in OHLC format, one unit of duration between successive intervals is effectively excluded. To illustrate, with 1-s observations, the fractional loss with per minute interval is $1/60 \approx 0.01667$ (or 1.667 %). The effective loss diminishes further as larger aggregation intervals are applied. For instance, using similar sampling frequency, 5 min', per hour and per day interval fractional losses are $1/300 \approx 0.00333$ (or 0.33 %), $1/3600 \approx 0.00028$ (or 0.028 %) and $1/86400 \approx 0.00001157$ (or 0.00001 %) respectively. Informed by recent advances in volatility modelling and time-series imputation [60–62], such a vanishingly small loss is well

within the class of negligible irregularities typically assumed to be “missing at random” and can therefore be safely ignored for distributional analysis and aggregation experiments. However, if the framework were scaled down to high-frequency or tick-level strategies where microstructure noise, order-book dynamics, and millisecond-level event timing materially influence trading outcomes, even small data gaps could interact with execution logic or short-horizon volatility estimates. While this does not affect the present analysis, it highlights the need for more stringent handling of any gaps when extending the framework to ultra-high-frequency contexts.

The 15,556,320 sequential prices presented in per hour OHLC format translates to 4320 h, while 31,112,640 and 46,668,960 sequential prices translate to 8640 and 12960 h respectively. Datasets were subjected to the same pre-processing pipeline to standardize frequency, impute missing values, normalize scales, and structure input-output windows for supervised learning. Our experiments are univariate, focusing on Close value.

3.2. Statistical diagnostics and data characterization

Prior to model training, a series of exploratory experiments were conducted to evaluate the presence of well-documented stylized facts in both the real and synthetic time series datasets. These diagnostics were essential for validating the structural properties of the data and assessing its suitability for time series modeling.

Non-stationarity occurs when statistical properties such as the mean or variance evolve over time, implying that the underlying distribution is unstable. To assess this property, the ADF test and the ACF were employed, both being fundamental tools in time series analysis [10]. The ADF test evaluates the presence of unit roots based on Equation (5):

$$x_t = \rho x_{t-1} + \epsilon_t \quad (5)$$

where x_t is series at time t , and ϵ is a white noise error term.

If $|\rho| < 1$, the series is stationary, if $\rho = 1$, the series follows a random walk, possesses a unit root, and is therefore non-stationary. Typical critical values for the ADF statistic for samples of 1000 or more observations are -3.43 (1 %), -2.86 (5 %), and -2.57 (10 %). If the computed ADF statistic is greater than the corresponding critical value (i.e., less negative), we fail to reject the null hypothesis of a unit root, indicating that the series is non-stationary. Equivalently, a p-value greater than 0.05 leads to the same conclusion, failure to reject the null hypothesis and confirmation of non-stationarity.

Such analysis on synthetic data evaluated whether essential temporal dependencies and non-stationary features observed in real forex data (e.g., evolving trends or autocorrelation) were faithfully replicated. A non-stationary series implies that shocks have lasting effects on future values, reflecting dynamic market behavior typical of real financial systems.

The autocorrelation function complements this analysis and is defined as (6):

$$\rho_k = \frac{\text{Cov}(x_t, x_{t-k})}{\text{Var}(x_t)} \quad (6)$$

where ρ_k represents the correlation between observations separated by k lags.

The ACF, computed up to 80 lags, visually reveals temporal dependencies, gradual decay indicates non-stationarity, while rapid decay suggests stationarity. The combined use of the ADF and ACF thus provided both statistical and visual confirmation of non-stationarity.

Volatility clustering, where high-volatility periods tend to follow high-volatility periods (and vice versa), was tested using the Ljung–Box test on squared returns [63]. Low p-values ($p < 0.05$) across lags 10–80 indicate significant autocorrelation in volatility, confirming clustering behavior. This is often modeled using a GARCH-type process as expressed in Equations (7) and (8):

$$r_t = \sigma_t \varepsilon_t, \varepsilon_t \sim N(0, 1) \quad (7)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \quad (8)$$

where r_t denotes returns, and σ_t^2 is the conditional variance capturing periods of high and low volatility.

Fat (heavy) tails, reflected by high kurtosis, indicate an increased probability of extreme returns. Kurtosis is given by Equation (9):

$$K = \frac{E[(r_t - \mu)^4]}{(\sigma^4)} \quad (9)$$

A normal distribution has $K = 3$; $K > 3$ implies leptokurtosis, consistent with extreme price movements common in financial markets.

The leverage effect describes the negative correlation between past returns and future volatility: This relationship is expressed in Equation (10):

$$\text{Corr}(r_{t-1}, \sigma_t^2) < 0 \quad (10)$$

This effect is examined by regressing squared returns on lagged returns using ordinary least squares (OLS). A significant negative coefficient ($p < 0.05$) confirm that negative shocks increase subsequent volatility, an essential market asymmetry reflecting investor risk responses.

These high level characterizations are used to inform and evaluate models of markets [7], and replicating these features in synthetic data enhances model realism while ensuring that ML systems trained on such data generalize better to actual market conditions [64].

3.3. Distance-based validation metrics

To assess the fidelity of synthetically generated forex time series relative to real market data, this study employs two complementary distance-based validation measures: the Wasserstein distance and the Hausdorff distance. These metrics capture distinct but interrelated aspects of similarity with Wasserstein distance addressing distributional realism and Hausdorff distance focusing on temporal coherence. The Wasserstein distance measures the minimal “cost” of transforming one probability distribution into another. Given the non-Gaussian, heavy-tailed nature of forex returns, this distance provides a robust basis for quantifying how closely the synthetic data replicates the statistical properties of real exchange rate movements. It is defined in (11):

$$W_p(P_X, P_Y) = \left(\inf_{\gamma \in \Gamma(P_X, P_Y)} \int_{\mathbb{R} \times \mathbb{R}} |x - y|^p d\gamma(x, y) \right)^{\frac{1}{p}} \quad (11)$$

where P_X and P_Y denote the empirical distributions of real and synthetic returns, and $\Gamma(P_X, P_Y)$ is the set of all admissible couplings with those marginals. Equation (11) represents the general formulation, which simplifies for one-dimensional series as shown in Equation (12):

$$W_p(P_X, P_Y) = \left(\int_0^1 |F_X^{-1}(u) - F_Y^{-1}(u)|^p du \right)^{\frac{1}{p}} \quad (12)$$

where F_X^{-1} and F_Y^{-1} represent their respective quantile functions.

The Wasserstein distance is therefore adopted as the primary validation metric, as it effectively captures statistical congruence between real and synthetic data distributions, including volatility clustering and tail behavior.

In contrast, the Hausdorff discrepancy evaluates the geometric similarity of price trajectories, emphasizing the largest minimal deviation between two time-series paths. For discrete forex series $X = \{x_t\}$ and $Y =$

$\{y_t\}$, it extends the general formulation in Equation (2) and is normalized for empirical application as shown in Equation (13):

$$HD(X, Y) = \frac{1}{T} \sum_{t=1}^T \min_s \|x_t - y_s\| \quad (13)$$

where T denotes the number of observations.

This measure functions as a complementary validation criterion, providing insight into the temporal and geometric coherence between real and synthetic price paths, including alignment of volatility bursts and directional trends.

Together, the Wasserstein distance and Hausdorff discrepancy form a dual validation framework that ensures both distributional fidelity by confirming that synthetic data replicate the empirical statistical features of forex markets and structural realism, by verifying that generated trajectories preserve plausible geometric and temporal patterns, thereby enabling comprehensive evaluation of synthetic forex time series across statistical and geometric dimensions.

3.4. Similarity measures

Real and synthetic OHLC datasets were aligned to a common feature space and preprocessed to ensure comparability. Each dataset was shuffled to reduce temporal bias, differenced to mitigate non-stationarity, and standardized using z-score normalization across all variables, including Close price. This produced scale-aligned, stationary inputs suitable for metric-based validation.

A four-part multi-metric framework was implemented to evaluate the degree of similarity between real and synthetic datasets. The Wasserstein-Cramér's V test quantified distributional alignment, while the Novelty Test (nearest-neighbor distance) assessed structural overlap with the real data manifold. The Domain Classifier Test used a Random Forest to estimate discriminative detectability, and the Anomaly Detection Test employed an Isolation Forest to evaluate internal coherence. Together, these tests capture complementary dimensions of similarity beyond conventional one-dimensional metrics.

Interpretation followed empirically derived benchmarks. Real vs real comparisons were first performed to establish baseline metric ranges, capturing natural variability in the data and the sensitivity of each similarity measure. Typical real vs real results include Wasserstein-Cramér's V ≈ 0.85 –1.00, novelty < 0.08 , classifier accuracy 0.45–0.60, and anomaly rate $< 10\%$. Real vs synthetic comparisons were then evaluated against these baselines, with acceptable fidelity ranges of V ≈ 0.65 –0.85, novelty 0.05–0.12, accuracy 0.55–0.75, and anomaly 10–25%. Metrics outside these bounds indicate potential structural or scaling deviations, guiding iterative refinement of the synthetic data generation process.

3.5. Predictive modeling

To evaluate predictive performance, ARIMA, XGBoost, LSTM, N-BEATS, and DLinear models were trained independently on real and synthetic datasets. All data were cleaned, sorted chronologically, and normalized using a standard scaler, with a fixed random seed (42) for reproducibility. Blocked cross-validation (70% contiguous training, 30% testing) was applied to preserve temporal dependencies. Sequential models (LSTM, N-BEATS, and DLinear) used overlapping input-output sequences with a window size of 20. LSTM employed two stacked layers (64 and 32 units) with dropout (0.2), ReLU activation, Adam optimizer ($\text{lr} = 1\text{e-}3$), batch size of 32, and early stopping over 30 epochs. N-BEATS consisted of three blocks with hidden size 128 and theta size 20, trained under the same optimizer and epochs. DLinear was configured with linear layers for trend and seasonal decomposition over

the input sequence, trained using the same optimizer and epochs. XGBoost used 300 trees, max depth 6, and learning rate 0.05, while ARIMA was configured as (2,1,2) to account for short-term dependencies and non-stationarity. Model performance was evaluated using MAE, MSE, R^2 , MAPE, SMAPE, and MASE, providing a comprehensive comparison of forecasting capability between real and synthetic data.

3.6. Residual-based diagnostics

In addition to conventional performance measures (MAE, R^2), residual-based diagnostics were conducted to assess dynamic fidelity. The ADF test examined residual stationarity, while the Ljung-Box test evaluated serial independence across lags. Residual behavior was compared for models trained on real and synthetic data to verify whether stochastic dependencies and noise structures were preserved in the generated series. Model residual definition is expressed in (14):

$$r_t = y_t - \hat{y}_t \quad (14)$$

where y_t is the observed price(ratio) (real or synthetic), and $\hat{y}_t$ is the model-predicted value at time t . Residual r_t capture the unexplained variation or stochastic component not modeled by the learning algorithm.

4. Results

This section presents the results of all experimental procedures, encompassing data generation, statistical validation through stylized facts, distance and similarity assessments, and ML model evaluations. The foundation of these experiments lies in the stochastic data generation process outlined in **Pseudocode 1** (Forex_Game_Simulator), which simulates trader-market interactions under varying risk conditions. The pseudocode produces sequences of trade types, risk appetites, and evolving ratio values that capture essential market dynamics. These outputs form the synthetic datasets subsequently analyzed to assess structural fidelity, distributional similarity, and predictive performance relative to real financial time series.

4.1. Generated data

Pseudocode 1 was implemented in Python to generate six datasets. **Table 1** presents a truncated subset of 4320 sample dataset, showing selected entries from the beginning and end of the sequence after pre-processing 15,556,320 price progression values of one of the simulations. **Fig. 4** illustrates the corresponding line graph of the **Close** values.

4.2. Stationarity and temporal dependency analysis

The analysis of the time series reveals several core stylized facts

Table 1

Structure of the preprocessed multivariate price progression dataset showing hourly OHLC evolution after aggregation and formatting.

SN	OPEN	HIGH	LOW	CLOSE
1	2.0828	2.08285	2.07356	2.07859
2	2.0787	2.0915	2.07835	2.09039
3	2.09027	2.09353	2.08597	2.09288
4	2.0929	2.12372	2.09213	2.09499
5	2.09485	2.10067	2.09169	2.0956
6	2.09546	2.09706	2.08543	2.09673
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
4318	2.13301	2.13621	2.13163	2.13248
4319	2.13264	2.13454	2.12777	2.1292
4320	2.12925	2.13669	2.12638	2.13522

typical of financial markets. The ADF test results (**Table 2**) show test statistics ranging from -1.01 to -2.84 , all above the 5 % critical threshold (approximately -2.86). This indicates a consistent failure to reject the null hypothesis of a unit root, confirming that all examined series are non-stationary and exhibit time-varying statistical properties.

To further investigate temporal dependencies, the Ljung-Box test on squared returns for all datasets yielded p-values < 0.001 across all lag ranges (10–80). Results for one of the experiments (both synthetic and real) for 4320 sample dataset is in **Table 3**. This confirms strong autocorrelation in volatility, indicative of volatility clustering, where high-volatility periods are followed by further turbulence.

Regression results show consistently positive and significant lagged-return coefficients (≈ 2.3 – 6.0 , $p < 0.001$) across all datasets, indicating pro-cyclical volatility rather than a negative leverage effect. This aligns with prior findings in foreign exchange markets, confirming that both real and synthetic data faithfully capture the symmetric volatility dynamics typical of FX returns.

Additionally, all datasets exhibited excess kurtosis (**Table 4**), with values well above the Gaussian benchmark of 3, confirming the presence of fat tails. This implies a higher-than-normal probability of extreme returns, a hallmark of real financial time series.

Collectively, these results confirm that both real and synthetic time series reproduce the key empirical regularities of financial data namely, non-stationarity, volatility clustering, and fat-tailed return distributions, while the leverage effect, as expected for foreign exchange markets, remains weak or statistically indistinct. Together, these properties constitute the foundational stylized facts underpinning market micro-structure modeling and serve as benchmarks for assessing the realism of synthetic data. **Fig. 3** illustrates the replication of key stylized facts in synthetic forex data compared to real series, highlighting similarities in volatility patterns, return distributions, and temporal correlations.

4.3. Distance metrics and similarity tests

To evaluate the fidelity of the synthetic data relative to real-world benchmarks, multiple distance and similarity measures were applied (**Tables 5 and 6**).

The Wasserstein-Cramér's V test produced values between 0.849 and 0.988, with the **Close** price consistently achieving excellent similarity (≥ 0.95), indicating that the synthetic data effectively captures the distributional shape of real price series. The Novelty Test, based on mean nearest-neighbor distances, yielded values ranging from 0.036 to 0.135, suggesting that synthetic sequences exhibit strong structural resemblance to real data with minimal divergence. The Domain Classifier Test reported accuracy levels between 0.61 and 0.79, implying that the classifier could not easily distinguish between real and synthetic samples, evidence of moderate detectability and hence high realism of the generated data. Furthermore, Anomaly Detection identified low anomaly rates (≈ 4 – 12 %) for larger datasets, indicating internal structural coherence and well-preserved feature relationships in synthetic series.

Complementary distance metrics reinforced these findings. The Hausdorff distance remained below 0.013 (raw scale) across all dataset sizes, while the Wasserstein distance stayed under 0.002. Scaled variants (3–5 for Hausdorff; 0.4–0.8 for Wasserstein) confirm low dissimilarity even after normalization.

These consistently low distance measures underscore the strong fidelity of synthetic data in reproducing the topological and statistical structure of real forex time series, making it suitable for downstream modeling tasks.

4.4. Predictive performance on real and synthetic datasets

The predictive utility of synthetic data was evaluated using five representative forecasting models: ARIMA, XGBoost, LSTM, N-BEATS and DLinear. Performance metrics including MAE, MSE, R^2 , MAPE, SMAPE, and MASE were computed across three dataset sizes (4320;

Table 2

ADF test results for real and synthetic datasets across 4,320, 8640 and 12,960 sample datasets.

Dataset Size	4320			8640			12960		
Dataset	Kaggle	ForexSB	Synthetic	ForexSB	Kaggle	Synthetic	Kaggle	ForexSB	Synthetic
ADF	−0.471	−1.295	−0.669	−1.558	−0.819	−0.986	−1.426	−1.646	−0.2079
p-value	0.898	0.6312	0.855	0.505	0.813	0.758	0.570	0.459	0.938

Table 3

Ljung–Box Test on Squared Returns for real and synthetic datasets across 4320 sample dataset.

Synthetic			Real		
Lag	Statistics	p-value	Lag	Statistics	p-value
10	168.497052	5.6784e-31	10	149.805710	4.0868e-27
20	187.852446	2.7983e-29	20	247.951969	2.9554e-41
30	195.382621	3.6066e-26	30	344.098136	4.7356e-55
40	202.334094	1.4571e-23	40	377.327968	1.8326e-56
50	229.036607	9.6945e-25	50	419.804168	6.7423e-60
60	241.927290	1.0819e-23	60	439.676518	3.6392e-59
70	253.819702	1.1646e-22	70	443.143275	1.3240e-55
80	272.840592	7.0568e-23	80	462.434209	3.5534e-55

8640; 12,960 sample dataset). Results are summarized in Tables 7–9 and compared across increasing data volumes for both real and synthetic series.

From Tables 7–9, XGBoost achieved consistently superior results across all datasets, with $R^2 \geq 0.98$ on large datasets and the lowest MAE, MSE, and MAPE values. This demonstrates its robustness in capturing nonlinear dependencies in both real and synthetic data. LSTM performance improved markedly with data size, reaching $R^2 > 0.90$ on large real datasets, highlighting its ability to model long-term temporal dependencies once sufficient training data are available. N-BEATS showed high stability and adaptability, particularly on synthetic datasets where R^2 ranged from 0.77 to 0.91. Its error metric MAE was consistently lower than those of ARIMA and LSTM, reflecting its proficiency in modeling both local and global time-series patterns. From Tables 7–9, DLinear consistently outperformed all other models, achieving the highest R^2 (up to 0.9988 on large synthetic datasets) and the lowest MAE, MSE, and MAPE values across all dataset sizes. Its linear decomposition approach

effectively captures trend and seasonal components while maintaining computational efficiency, making it particularly well-suited for both real and synthetic forex data. ARIMA, by contrast, underperformed significantly, with large errors and negative R^2 values, underscoring the inadequacy of linear models in capturing the nonlinear and non-stationary behavior of forex data.

To ensure comparability across scales, the Mean Absolute Scaled Error (MASE) was used as an additional indicator of predictive accuracy. DLinear achieved the lowest MASE values across nearly all configurations, followed closely by XGBoost and N-BEATS, reaffirming their superior generalization capabilities. LSTM demonstrated steady improvement in MASE with increasing dataset size, confirming its scalability, while ARIMA consistently recorded the highest MASE across all configurations.

4.5. Residual dynamics of ML models

Table 10 reports residual diagnostics for real and synthetic datasets across 12,960 samples, evaluated using the ADF and Ljung–Box tests. “Stationary” denotes rejection of the unit root hypothesis ($p < 0.05$), while Ljung–Box statistics quantify serial dependence across lags.

XGBoost produced stationary residuals with low autocorrelation for both datasets, indicating effective noise modeling. N-BEATS exhibited partial stationarity and reduced autocorrelation relative to ARIMA and LSTM, whose residuals remained largely non-stationary and serially correlated. Rather than reflecting simple model failure, the persistent residual structure observed in ARIMA and LSTM reinforces the realism and inherent complexity of the synthetic series, which preserves nonlinear dependencies and regime dynamics that simpler or more rigid models struggle to fully capture. This behavior underscores that the synthetic data retains meaningful structural difficulty comparable to

Table 4

Kurtosis and Kurtosis of Returns for real and synthetic datasets across 4,320, 8640 and 12,960 sample dataset.

Dataset Size	4320			8640			12960		
Dataset	Kaggle	ForexSB	Synthetic	ForexSB	Kaggle	Synthetic	Kaggle	ForexSB	Synthetic
Kurtosis	8.520	8.229	3.446	3.539	7.64	3.267	4.097	4.154	3.574
Kurtosis of Returns	7.54	6.93	1.27	4.96	6.728	1.18	1.48	4.83	1.20

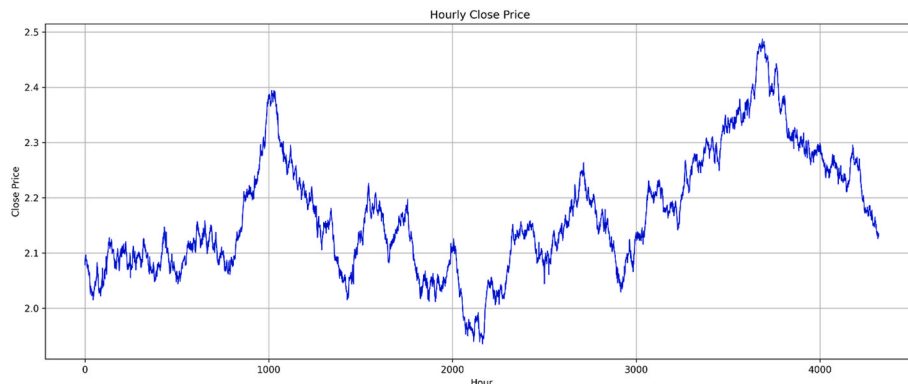


Fig. 3. Stylized facts comparison between real (left) and synthetic (right) forex data: volatility clustering (top), return distributions (middle), and autocorrelation of absolute returns (bottom).

Stylized Facts: Real vs Synthetic Forex Data (8640 Observations)

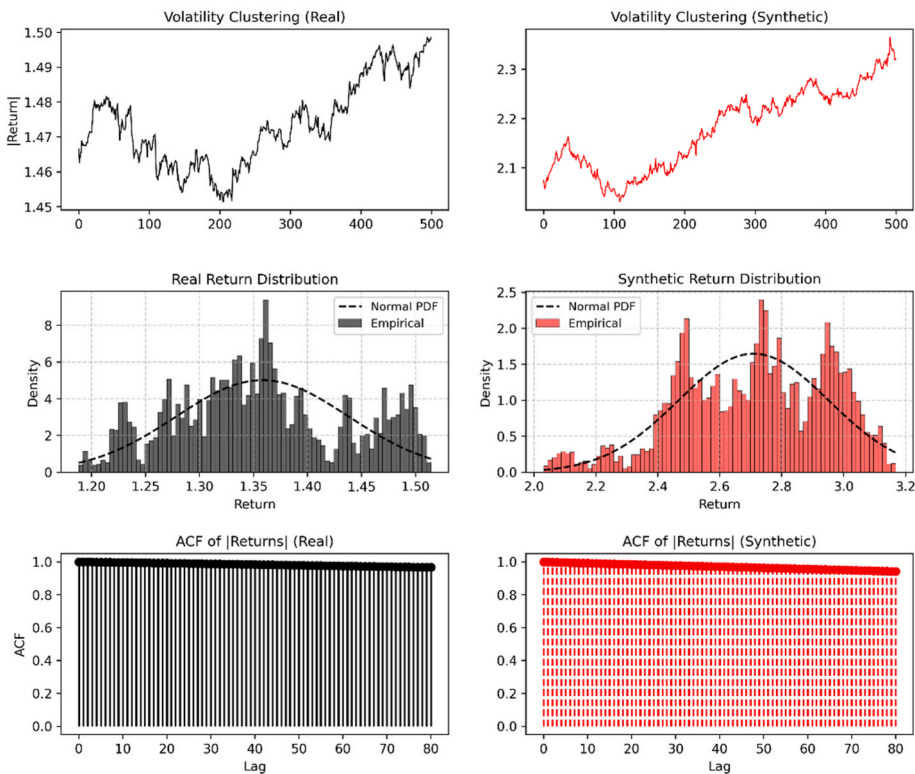


Fig. 4. Line plot of close prices over time from a simulated OHLC sequence, illustrating the temporal dynamics of the generated series.

Table 5
Results of similarity and structural consistency tests, including Wasserstein Cramér's V, novelty, domain classifier, and anomaly measures for real and synthetic datasets across multiple data sizes.

Test	4320 sample dataset		8640 sample dataset		12960 sample dataset	
	ForexSB	Kaggle	ForexSB	Kaggle	ForexSB	Kaggle
Wasserstein-Cramér's V- Close Price	0.925-Excellent similarity	0.934-Excellent similarity	0.979-Excellent similarity	0.937-Excellent similarity	0.988-Excellent similarity	0.971-Excellent similarity
Novelty Test	0.0957-Moderately realistic: small differences exist.	0.1355-Noticeable structural gap between synthetic and real.	0.0364-Very realistic: synthetic data closely matches real patterns.	0.0730-Moderately realistic: small differences exist.	0.0374-Very realistic: synthetic data closely matches real patterns.	0.0549-Moderately realistic: small differences exist.
Domain Classifier	0.79-Highly detectable: synthetic data diverges strongly from real.	0.77-Highly detectable: synthetic data diverges strongly from real.	0.67-Moderate detectability: synthetic data has some differences.	0.76-Highly detectable: synthetic data diverges strongly from real	0.61-Moderate detectability: synthetic data has some differences.	0.71-Moderate detectability: synthetic data has some differences.
Anomaly Detection	28.19 %-Some mismatches in structure detected.	24.49 %-Some mismatches in structure detected	4.59 %-Synthetic data structure is consistent with real.	23.04 %-Some mismatches in structure detected.	6.81 %-Synthetic data structure is consistent with real.	11.60 %-Some mismatches in structure detected.

Table 6
Results of geometric and distributional similarity tests based on Wasserstein and Hausdorff distances for real and synthetic datasets across multiple data sizes.

Test	4320 sample dataset				8640 sample dataset				12960 sample dataset			
	ForexSB		Kaggle		ForexSB		Kaggle		ForexSB		Kaggle	
	Raw	Scaled	Raw	Scaled	Raw	Scaled	Raw	Scaled	Raw	Scaled	Raw	Scaled
Hausdorff distance	0.0074	3.2013	0.0079	3.5595	0.0093	3.6580	0.012	4.702	0.0093	3.6944	0.0115	4.7966
Wasserstein Distance	0.0013	0.5760	0.0016	0.6992	0.0016	0.6114	0.0019	0.785	0.0015	0.5968	0.0019	0.805

Table 7

Performance evaluation of ARIMA, LSTM, XGBoost, N-BEATS and DLinear models on 4,320 sample dataset using MAE, MSE, R^2 , MAPE, SMAPE, and MASE metrics (Synthetic vs ForexSB).

Model	Dataset	MAE	MSE	R2	MAPE	SMAPE	MASE
ARIMA	Real	0.0573	0.0055	-1.4136	4.5127	4.3439	37.0003
XGBoost	Real	0.0424	0.0039	-0.7079	3.3710	3.2503	27.3632
LSTM	Real	0.0648	0.0064	-1.8127	5.0827	4.8878	41.7795
NBEATS	Real	0.0252	0.0009	0.5914	1.9730	1.9439	16.2525
DLinear	Real	0.0108	0.0002	0.9200	0.8473	0.8416	6.9680
ARIMA	Synthetic	0.1549	0.0313	-0.8551	6.0355	6.2492	31.2130
XGBoost	Synthetic	0.0239	0.0016	0.9058	0.9142	0.9252	4.8272
LSTM	Synthetic	0.1085	0.0169	-0.0001	4.4044	4.3519	21.8966
NBEATS	Synthetic	0.0301	0.0012	0.9274	1.1742	1.1823	6.0656
DLinear	Synthetic	0.0118	0.0002	0.9867	0.4772	0.4783	2.3712

Table 8

Performance evaluation of ARIMA, LSTM, XGBoost, N-BEATS and DLinear models on 8640 sample dataset using MAE, MSE, R^2 , MAPE, SMAPE, and MASE metrics (Synthetic vs Kaggle).

Model	Dataset	MAE	MSE	R2	MAPE	SMAPE	MASE
ARIMA	Real	0.0776	0.0073	-4.8071	5.6919	5.8920	55.2359
XGBoost	Real	0.0018	0.0000	0.9948	0.1348	0.1348	1.3004
LSTM	Real	0.0976	0.0108	-7.6699	7.1764	7.4762	69.5864
NBEATS	Real	0.0172	0.0004	0.6524	1.2596	1.2708	12.2975
DLinear	Real	0.0039	0.0000	0.9813	0.2883	0.2877	2.8038
ARIMA	Synthetic	0.1149	0.0190	-0.0087	3.8561	3.8754	23.1254
XGBoost	Synthetic	0.0648	0.0116	0.3870	2.0747	2.1345	13.0267
LSTM	Synthetic	0.1361	0.0303	-0.6040	4.4588	4.6074	27.3690
NBEATS	Synthetic	0.0294	0.0012	0.9374	1.0058	1.0017	5.9102
DLinear	Synthetic	0.0165	0.0004	0.9768	0.5424	0.5444	3.3151

Table 9

Performance evaluation of ARIMA, LSTM, XGBoost, N-BEATS and DLinear models on 12960 sample dataset using MAE, MSE, R^2 , MAPE, SMAPE, and MASE metrics (Synthetic vs Kaggle).

Model	Dataset	MAE	MSE	R2	MAPE	SMAPE	MASE
ARIMA	Real	0.0321	0.0015	-1.9398	2.8089	2.7503	57.4326
XGBoost	Real	0.0008	0.0000	0.9970	0.0725	0.0725	1.4997
LSTM	Real	0.0419	0.0023	-3.3442	3.6598	3.5740	75.1013
NBEATS	Real	0.0220	0.0007	-0.3677	1.9273	1.8999	39.4356
DLinear	Real	0.0015	0.0000	0.9934	0.1267	0.1265	2.5980
ARIMA	Synthetic	0.3222	0.1377	-1.7837	9.3077	9.8940	64.1437
XGBoost	Synthetic	0.1802	0.0514	-0.0411	5.1401	5.3554	35.8726
LSTM	Synthetic	0.2918	0.1075	-1.1768	8.4779	8.9073	58.0928
NBEATS	Synthetic	0.0265	0.0009	0.9810	0.7692	0.7729	5.2710
DLinear	Synthetic	0.0062	0.0001	0.9988	0.1854	0.1852	1.2314

real-world forex environments. The superior residual behavior of XGBoost and N-BEATS is consistent with their earlier performance in distance, similarity, and predictive analyses, confirming their fidelity to real market dynamics.

Across 4,320, 8640 and 12,960 sample experiments, residual stability improved with increasing sample size, a clear size-dependent enhancement in model consistency.

4.6. Empirical validation of synthetic data fidelity and utility

Across all experimental tiers, statistical characterization, similarity evaluation, predictive modeling and residual dynamics, the synthetic data faithfully reproduce the empirical and predictive structure of real forex time series. Key stylized facts, including non-stationarity, volatility clustering, and fat-tailed return distributions, were successfully replicated, while the leverage effect remained weak, consistent with typical FX market behavior. Similarity metrics confirm strong distributional, structural, and topological resemblance to real data. Predictive models (particularly XGBoost, LSTM, N-BEATS and DLinear) trained on synthetic data exhibited performance metrics comparable to those trained on real datasets and residual dynamics were consistent.

These results collectively demonstrate that the synthetic datasets preserve essential statistical and temporal dynamics, enabling credible predictive modeling under data constraints. The validation framework thus confirms both the fidelity and utility of the synthetic data for robust ML experimentation in forex forecasting.

5. Discussion

This study systematically evaluated synthetic OHLC forex datasets against real data using a multi-metric validation framework that integrates model-based validation, cross-model generalizability, and stylized fact reproduction. Predictive performance was assessed using ARIMA, XGBoost, LSTM, N-BEATS, and DLinear, providing a comprehensive perspective on both the statistical fidelity and forecasting utility of the synthetic sequences.

5.1. Stationarity and autocorrelation

The ADF tests indicate that all datasets, real and synthetic, are non-stationary across sample sizes, consistent with typical forex time series behavior. The Ljung-Box test on squared returns confirmed significant

Table 10
Residual diagnostic results for each model across real and synthetic datasets.

Model	Dataset	Lags Range	LB Statistics Range	p-Value	Stationary
XGBoost	Real	5–20	5689–24886	0	YES
XGBoost	Synthetic	5–20	61–74247	0	YES
DLinear	Real	5–20	34778–39922	0	YES
DLinear	Synthetic	5–20	7021–21740	0	YES
N-BEATS	Real	5–20	24111–76015	0	NO
N-BEATS	Synthetic	5–20	5211–60339	0	YES
LSTM	Real	5–20	24763–76016	0	NO
LSTM	Synthetic	5–20	24126–75480	0	NO
ARIMA	Real	5–20	24899–76136	0	NO
ARIMA	Synthetic	5–20	24249–75581	0	NO

autocorrelation in both domains, with slightly lower statistics for synthetic sequences, suggesting that volatility clustering is preserved, albeit with minor smoothing of extreme dependencies. Stylized fact visualizations corroborate these findings, showing that synthetic data captures key features such as clustered volatility and the general shape of return distributions.

5.2. Higher-order moments

Analysis of kurtosis and return distributions indicates that the synthetic data successfully reproduces the heavy-tailed behavior characteristic of forex markets, although smaller sample sizes show moderately attenuated extremes. This suggests that the generative mechanism captures the broader distributional structure while slightly under-representing the most severe tail events, a point of relevance for risk-sensitive modeling. While this limitation is acknowledged, future iterations of the simulator, particularly those incorporating adaptive agent behaviors or more dynamically responsive market shocks may help better emulate rare, high-impact deviations without compromising the underlying behavioral and game-theoretic framework.

5.3. Structural and distributional consistency

Distributional similarity and structural consistency were assessed via Wasserstein-Cramér’s V, novelty tests, domain classifiers, anomaly detection, and geometric measures (Hausdorff and Wasserstein distances). High similarity scores (>0.92) and low geometric distances confirm that synthetic datasets closely replicate the structural and distributional characteristics of real data. Minor deviations, particularly

in smaller datasets, diminish with increasing sample size, indicating the robustness of the generative approach.

5.4. Predictive performance across models

Performance evaluation demonstrates that DLinear consistently achieves the lowest error metrics (MAE, MSE, MASE) across all datasets, highlighting its ability to disentangle linear trend and seasonal components from nonlinear effects. N-BEATS exhibits strong predictive capability, particularly on synthetic data, while XGBoost provides robust, scale-independent performance, effectively capturing nonlinear dependencies. LSTM shows sensitivity to dataset size, struggling with smaller synthetic sequences, and ARIMA is constrained by its linear assumptions and short-term dependency modeling. Importantly, predictive patterns on synthetic data mirror those observed on real data, confirming that synthetic sequences preserve essential forecasting signals.

5.5. Residual diagnostics

Residual analysis reinforces these conclusions: DLinear and XGBoost produce stationary residuals across all datasets, indicating accurate modeling of temporal dependencies. In contrast, residuals for LSTM, N-BEATS, and ARIMA occasionally retain autocorrelation, suggesting incomplete capture of stochastic complexity or overfitting in smaller datasets.

5.6. Stylized fact reproduction and multi-metric validation

Synthetic series reproduce key stylized facts, including volatility clustering, leptokurtic returns, non-stationarity, and minimal autocorrelation in raw returns. Complementary metrics such as Wasserstein-Cramér’s V, novelty, domain classifier, anomaly detection, and geometric distances confirm both distributional and structural fidelity. The inclusion of agent-driven trader-market interactions in the generative process further enhances behavioral realism, extending the framework beyond purely statistical replication.

5.7. Discussion summary

Overall, synthetic forex datasets consistently replicate the distributional and predictive patterns of real data across classical, ensemble, and deep learning models. DLinear serves as a linear decomposition baseline, disentangling trend-seasonal effects from nonlinear dependencies and enhancing interpretability. Minor deviations in higher-order moments

Table 11
Discussion summary table.

Aspect	Metric/Test	Real Data	Synthetic Data	Key Observation
Stationarity	ADF test	Non-stationary	Non-stationary	Synthetic data preserves non-stationarity across sample sizes
Autocorrelation	Ljung–Box on squared returns	Significant autocorrelation	Significant, slightly lower	Volatility clustering reproduced but slightly smoothed
Return Distribution	Kurtosis/Kurtosis of Returns	Heavy tails, kurtosis >4	Heavy tails, slightly lower kurtosis	Synthetic data captures distribution shape, minor underrepresentation of extremes
Distributional Similarity	Wasserstein-Cramér’s V	>0.92	>0.92	Excellent similarity in price distributions
Structural Gap	Novelty/Domain Classifier/Anomaly Detection	Minor deviations detected	Minor deviations detected	Small structural gaps reduce with larger datasets
Geometric Similarity	Hausdorff/Wasserstein distance	Low distances	Low distances	Distributional geometry well-replicated
Forecasting (MAE, MSE, MASE)	ARIMA	Moderate errors	Higher errors	ARIMA limited by short-term assumptions
	XGBoost	Low errors	Low errors	Strong performance, captures nonlinear patterns
	LSTM	Moderate-high errors	Moderate errors	Sensitive to dataset size, struggles with small synthetic data
	N-BEATS	Low errors	Low errors	Block-based architecture effective for sequence modeling
	DLinear	Lowest errors	Lowest errors	Linear decomposition baseline effectively captures trend-seasonal effects; improves interpretability
Residual Diagnostics	LB Statistics	Stationarity varies	Stationarity varies	DLinear & XGBoost residuals stationary; others show occasional non-stationarity

and anomaly measures indicate areas for future refinement to better capture extreme events. Overall, multi-metric evaluation demonstrates that properly validated synthetic data provide both reliable forecasting performance and meaningful insights into underlying market dynamics, supporting applications from model training to risk assessment and policy simulation. To address the possibility that the synthetic series may be overly smooth and therefore easier for linear models such as DLinear to predict, the simulator can be refined by introducing shorter and more frequent engineered volatility events. Real forex markets exhibit volatility clustering and rapid micro-regime shifts, and incorporating such fragmented shocks helps break overly deterministic trend-seasonal structure while preserving the behavioral logic of the game-theoretic design. Importantly, these micro-bursts also strengthen key stylized facts by producing more pronounced fat tails: irregular, high-impact deviations occurring at unpredictable intervals naturally increase kurtosis and tail thickness without resorting to unstructured noise. Together, these adjustments enhance the stochastic fidelity of the synthetic data and yield predictive dynamics that more closely resemble real-world market behavior. Table 11 synthesizes these findings by comparing real and synthetic datasets across statistical properties, structural similarity, and model-based forecasting performance, offering a consolidated view of their alignment and remaining gaps.

6. Conclusion

This study establishes a model-based validation framework for assessing the predictive equivalence of synthetic and real forex datasets, leveraging forecasting performance across ARIMA, XGBoost, LSTM, N-BEATS, and DLinear models. Synthetic data generated from a stochastic, game-theoretic foundation sustain strong predictive accuracy across linear and nonlinear paradigms, reproducing key stylized facts including volatility clustering, leptokurtic return distributions, non-stationarity, and weak leverage effects. Similarity metrics such as Wasserstein-Cramér's V, Hausdorff distances, and novelty tests confirm close alignment with real data, while residual diagnostics indicate that temporal dependencies and volatility structures are faithfully captured. DLinear provides a critical baseline for disentangling trend-seasonal from nonlinear components, enhancing interpretability and insight into the generative process. Across models and sample sizes, XGBoost and N-BEATS achieve the lowest error metrics and highest R^2 values, with LSTM performance improving for larger datasets, demonstrating robustness and cross-model generalizability. Minor deviations in higher-order moments and anomaly measures suggest underrepresentation of extreme events, highlighting opportunities for incorporating higher-frequency signals, adaptive agent behaviors, and tail-event modeling in future work. Collectively, these results establish that multi-metric validated synthetic datasets not only replicate statistical and predictive structures of real markets but also provide interpretable, reliable substitutes suitable for forecasting, backtesting, policy simulation, and risk assessment under constrained or privacy-sensitive data conditions.

CRediT authorship contribution statement

Patrick Naivasha: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **George Musumba:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Patrick Gikunda:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **John Wandeto:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Synthetic data was generated using method outlined in this paper while real forex data was retrieved from website listed under the reference section.

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